

- 实验报告
 - 数据类型
 - 1. int.m
 - 2. char.m
 - 3. char-int.m
 - 4. ptr-int.m
 - 5. ptr-char.m
 - 6. ptr-char-int.m
 - 7. arr.m
 - 8. arr-while.m
 - 9. arr-struct.m
 - 10. struct.m
 - 11. struct-arr.m
 - 12. struct-ptr.m
 - 控制结构
 - 13. if.m
 - 14. switch.m
 - 15. while.m
 - 16. while-bc.m
 - 17. for.m
 - 18. for-bc.m
 - 函数
 - 19. func-int.m
 - 20. func-char.m
 - 效率测试
 - loop.m

实验报告

数据类型

1. int.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini int.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm int.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine int.o
1
111-9-270

```

```

-----
CLOCK CYCLES : 147
MUL DIV : 1
MEM READ : 6
MEM WRITE : 6
-----

```

2. char.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini char.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm char.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine char.o

```

```

A C
CbcA
D

```

```

-----
CLOCK CYCLES : 180
MUL DIV : 0
MEM READ : 8
MEM WRITE : 7
-----

```

3. char-int.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini char-int.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm char-int.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine char-int.o

```

```

D A
AbcD
65989968

```

```

-----
CLOCK CYCLES : 258
MUL DIV : 0
MEM READ : 12
MEM WRITE : 10
-----

```

4. ptr-int.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object ? main ± ./mini ptr-int.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object ? main ± ./asm ptr-int.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object ? main ± ./machine ptr-int.o
7
717-3-90
111222

```

```

-----
CLOCK CYCLES : 229
MUL DIV : 1
MEM READ : 9
MEM WRITE : 10
-----

```

5. ptr-char.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object ? main ± ./asm ptr-char.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object ? main ± ./machine ptr-char.o

```

```

D A
AbcD
AB

```

```

-----
CLOCK CYCLES : 246
MUL DIV : 0
MEM READ : 11
MEM WRITE : 10
-----

```

6. ptr-char-int.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object ? main ± ./mini ptr-char-int.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object ? main ± ./asm ptr-char-int.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object ? main ± ./machine ptr-char-int.o

```

```

A 5
A5

```

```

-----
CLOCK CYCLES : 197
MUL DIV : 0
MEM READ : 9
MEM WRITE : 8
-----

```

7. arr.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini arr.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm arr.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine arr.o
5
17

```

```

-----
CLOCK CYCLES : 680
MUL DIV : 0
MEM READ : 21
MEM WRITE : 41
-----

```

8. arr-while.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini arr-while.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm arr-while.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine arr-while.o
5
141312111098765

```

```

-----
CLOCK CYCLES : 2893
MUL DIV : 20
MEM READ : 105
MEM WRITE : 136
-----

```

9. arr-struct.m

```

X zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini arr-struct.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm arr-struct.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine arr-struct.o
12ab

```

```

-----
CLOCK CYCLES : 1243
MUL DIV : 0
MEM READ : 32
MEM WRITE : 80
-----

```

10. struct.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini struct.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm struct.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine struct.o
5 6 7
107206305

-----
CLOCK CYCLES : 460
MUL DIV : 0
MEM READ : 14
MEM WRITE : 26
-----

```

11. struct-arr.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini struct-arr.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm struct-arr.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine struct-arr.o
12ab

-----
CLOCK CYCLES : 987
MUL DIV : 0
MEM READ : 24
MEM WRITE : 64
-----

```

12. struct-ptr.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini struct-ptr.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm struct-ptr.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine struct-ptr.o
999

-----
CLOCK CYCLES : 394
MUL DIV : 0
MEM READ : 9
MEM WRITE : 26
-----

```

控制结构

13. if.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini if.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm if.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine if.o
5 7
not equal
12

-----
CLOCK CYCLES : 146
MUL DIV : 0
MEM READ : 6
MEM WRITE : 6
-----

```

14. switch.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini switch.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm switch.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine switch.o
5
not 1 2 3

-----
CLOCK CYCLES : 146
MUL DIV : 0
MEM READ : 5
MEM WRITE : 7
-----

```

15. while.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini while.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm while.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine while.o
5
01234

-----
CLOCK CYCLES : 429
MUL DIV : 0
MEM READ : 20
MEM WRITE : 15
-----

```

16. while-bc.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini while-bc.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm while-bc.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine while-bc.o
15
012345678910break
11121314

```

```

-----
CLOCK CYCLES : 1353
MUL DIV : 0
MEM READ : 50
MEM WRITE : 50
-----

```

17. for.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini for.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm for.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine for.o
5
01234

```

```

-----
CLOCK CYCLES : 479
MUL DIV : 0
MEM READ : 25
MEM WRITE : 15
-----

```

18. for-bc.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine for-bc.o
15
012345678910break
0123456789continue11121314

```

```

-----
CLOCK CYCLES : 2844
MUL DIV : 0
MEM READ : 137
MEM WRITE : 86
-----

```

函数

19. func-int.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini func-int.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm func-int.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine func-int.o
5 8
8 9

-----
CLOCK CYCLES : 288
MUL DIV : 0
MEM READ : 12
MEM WRITE : 13
-----

```

20. func-char.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini func-char.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm func-char.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine func-char.o
A B
B C

-----
CLOCK CYCLES : 288
MUL DIV : 0
MEM READ : 12
MEM WRITE : 13
-----

```

效率测试

loop.m

```

zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./mini loop.m
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./asm loop.s
zjlhxg@KJ01 /mnt/c/Users/26937/Desktop/bianiyi/mini-object } main ± ./machine loop.o
1
2
3
5987654321 4987654321 3987654321 2987654321 1987654321
14 15

-----
CLOCK CYCLES : 3719
MUL DIV : 3
MEM READ : 178
MEM WRITE : 127
-----

```